

Chapter 4

Nominal and Effective Interest Rates

In the previous chapter, we established equivalence factors assuming that the interest is compounded once a year. However, in practical scenarios, interest is often compounded more frequently (e.g., daily, weekly, monthly, quarterly, or continuously). Because of this more frequent compounding, the actual interest accumulated over a year will be greater than what a simple stated rate implies. Therefore, we must formally distinguish between the stated (*nominal*) interest rate and the true (*effective*) interest rate.

Example 4.1. Consider a principal of \$100 to be invested for one year at an **interest rate** of 1% per month.

Calculate the amount of interest and the annual interest rate.

Answer

-

$$\begin{aligned} F &= P(1 + i)^n \\ &= 100(1.01)^{12} \\ &= \$112.68 \end{aligned}$$

The interest earned is \$12.68.

- The actual annual interest rate is $12.68/100 = 12.68\%$.

We say that the **effective** annual interest rate is 12.68%, and the **nominal** interest rate is 12%.

4.1 Nominal Interest Rate

Definition – Nominal Interest Rate r (APR)

A nominal interest rate r (often referred to in consumer finance as the Annual Percentage Rate or APR) is a rate that does not consider compounding.

$$r = \text{interest rate per period} \times \text{number of periods} \quad (4.1)$$

A nominal rate may be calculated for any time period longer than the time period stated by using Equation 4.1. For example, an interest rate of 1.5% per month corresponds to the following nominal rates, as shown in Table 4.1.

Time Period	Nominal Rate by Equation 4.1	What This Is
24 months	$1.5 \times 24 = 36\%$	Nominal rate per 2 years
12 months	$1.5 \times 12 = 18\%$	Nominal rate per 1 year
6 months	$1.5 \times 6 = 9\%$	Nominal rate per 6 months
3 months	$1.5 \times 3 = 4.5\%$	Nominal rate per 3 months

Table 4.1: Calculated nominal interest rates for various time periods

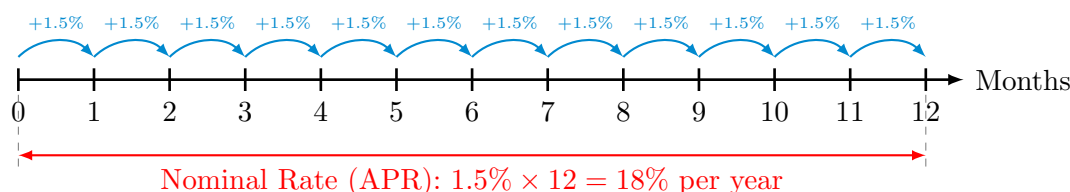


Figure 4.1: Visualizing the difference between actual compounding events (1.5% monthly hits) and the stated nominal annual rate over a 12-month period.

Note – Annual Percentage Rate (APR)

Nominal interest rate is commonly referred to as Annual Percentage Rate “APR”. While nominal rates can theoretically be calculated for any duration, in practice and in most problems in this course, “nominal rate” assumes an annual period (APR).

4.2 Effective Interest Rate

Definition – Effective Interest Rate

An effective interest rate i is a rate that does consider compounding.

The effective interest rate is the **actual** or observed rate that applies over a given period of time. In financial contracts, interest rates are frequently stated in the following format:

r (per year), compounded every CP

where CP is the compounding period (e.g., yearly, monthly, weekly) From this statement, we must calculate the actual effective interest rate that applies to each compounding period.

Note – Annual Percentage Yield (APY)

The effective interest rate i is called the Annual Percentage Yield “APY”.

4.2.1 Computing the Effective Interest Rate

Given that the effective interest rate is the actual rate in the compounding period, then

$$i = \frac{r}{m}$$

where $m = 1/CP$ is the compounding frequency per year¹. For instance, a nominal interest rate of 12% ($r = 12\%$) compounded monthly ($CP=1/12$ and $m = 12$) translates into an effective rate of 1% per month.

Let i_a be the effective annual rate. By setting the future worth F calculated with the effective annual rate equal to the future worth calculated with the compounded nominal rate, we can isolate i_a :

$$\begin{aligned} F &= P(1 + i_a) \\ &= P \left(1 + \frac{r}{m}\right)^m \end{aligned}$$

As a result, we can compute i_a as follows:

$$\begin{aligned} P(1 + i_a) &= P \left(1 + \frac{r}{m}\right)^m \\ 1 + i_a &= \left(1 + \frac{r}{m}\right)^m \\ i_a &= \left(1 + \frac{r}{m}\right)^m - 1 \end{aligned}$$

Definition – Effective Annual Interest Rate Formula

The effective annual interest rate is:

$$i_a = \left(1 + \frac{r}{m}\right)^m - 1$$

¹Given that CP is a fraction of a year.

4.3 Factors under m-time-a-year Compounding

In the previous chapter, we discussed engineering economy factors. These factors used the interest rate i per compounding period, that is r/m . When expressed with r and m , the Future Worth becomes:

$$F = P(1 + r/m)^k$$

where k is the total number of periods. Hence, the Future Worth Factor F/P can be written as:

$$(F/P, r, m, k) = (1 + r/m)^k$$

and $(F/P, r, m, k) \iff (F/P, i, k)$.

Similarly, the Present Worth factor P/F can be written using the nominal rate and compounding frequency as follows:

$$(P/F, r, m, k) = (1 + r/m)^{-k}$$

and $(P/F, r, m, k) \iff (P/F, i, k)$.

4.4 Continuous Compounding

If we allow compounding to occur more and more frequently, the compounding period (CP) becomes shorter and shorter and m , the number of times interest is compounded per year, becomes infinite. Let $CP \rightarrow 0$. Then $m = 1/CP \rightarrow \infty$. This situation is known as **continuous compounding**².

Definition – Continuous Compounding

Continuous compounding is present when the duration of CP, the compounding period, becomes infinitely small and m , the number of times interest is compounded per period, becomes infinite.

As compounding becomes continuous, we rely on the natural logarithm base e to model continuous growth. Given the definition of Euler's number e :

$$\lim_{h \rightarrow \infty} \left(1 + \frac{1}{h}\right)^h = e$$

²Continuous compounding appears in options pricing and certain financial models, but for physical engineering projects, monthly or quarterly compounding is the practical limit.

we can write the effective annual interest rate as:

$$\begin{aligned}\lim_{m \rightarrow \infty} i &= \lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^m - 1 \\ &= \lim_{h \rightarrow \infty} \left(1 + \frac{1}{h}\right)^{hr} - 1 \\ &= e^r - 1\end{aligned}$$

by taking $r/m = 1/h$ and consequently $m = hr$.

Definition – Effective Rate Under Continuous Compounding

Given a nominal annual rate $r\%$, the effective annual interest rate is:

$$i = e^r - 1$$

Important – Growth Factor Under Continuous Compounding

Under continuous compounding, the rate of growth of an investment over a time t expressed in years, i.e., the F/P factor, is

$$\lim_{m \rightarrow \infty} (1 + r/m)^{mt} = e^{rt}$$

4.4.1 Factors

Since $i = e^r - 1$, we can substitute i in the factors developed in Chapter 2 to find their corresponding continuous compounding factors.

For instance, the continuous counterpart of $P/F = (1 + i)^{-n}$ under continuous compounding is:

$$P/F = e^{-rt}$$

Note – Time Variable Notation

To keep discrete and continuous time different, we use t for continuous time, and n for discrete time.

A list of factors and the corresponding equations is shown in Table 4.2.

Example 4.2. You invest \$1 at a nominal rate of 8%. What would be the impact of different compounding periods ranging from yearly to continuously?

Name	Notation	Equation
Compound Amount	$(F/P, r, t)$	e^{rt}
Present Worth	$(P/F, r, t)$	e^{-rt}
Uniform Series Present Worth	$(P/A, r, t)$	$\frac{e^{rt}-1}{e^{rt}(e^r-1)}$
Uniform Series Compound Amount	$(F/A, r, t)$	$\frac{e^{rt}-1}{e^r-1}$
Capital Recovery	$(A/P, r, t)$	$\frac{e^{rt}(e^r-1)}{e^{rt}-1}$
Sinking Fund	$(A/F, r, t)$	$\frac{e^r-1}{e^{rt}-1}$

Table 4.2: Summary of Engineering Economy Factors for Continuous Compounding. r , nominal annual interest rate, compounded continuously; t , number of periods (years); A , annual equivalent amount (**occurs at the end of each year**); F , future equivalent; P , present equivalent.

Year	1	2	4	12	52	365	Continuous
1	\$1.0800	\$1.0816	\$1.0824	\$1.0830	\$1.0832	\$1.0833	\$1.0833
2	\$1.1664	\$1.1699	\$1.1717	\$1.1729	\$1.1734	\$1.1735	\$1.1735
3	\$1.2597	\$1.2653	\$1.2682	\$1.2702	\$1.2710	\$1.2712	\$1.2712
4	\$1.3605	\$1.3686	\$1.3728	\$1.3757	\$1.3768	\$1.3771	\$1.3771
5	\$1.4693	\$1.4802	\$1.4859	\$1.4898	\$1.4914	\$1.4918	\$1.4918
6	\$1.5869	\$1.6010	\$1.6084	\$1.6135	\$1.6155	\$1.6160	\$1.6161
7	\$1.7138	\$1.7317	\$1.7410	\$1.7474	\$1.7499	\$1.7506	\$1.7507
8	\$1.8509	\$1.8730	\$1.8845	\$1.8925	\$1.8955	\$1.8963	\$1.8965
9	\$1.9990	\$2.0258	\$2.0399	\$2.0495	\$2.0533	\$2.0543	\$2.0544
10	\$2.1589	\$2.1911	\$2.2080	\$2.2196	\$2.2242	\$2.2253	\$2.2255

Table 4.3: Future value compounding over different frequencies

Note – Excel NOMINAL and EFFECT Functions

Excel is also fitted with functions to compute the nominal and effective interest rates. The functions are *NOMINAL* and *EFFECT*. It will be left to you to explore these functions, or even use them to double-check your work.

4.5 Bonds

Definition – Bond

A bond is a loan or an IOU (I Owe You). A bond is a long-term note issued by a corporation or a government entity (the borrower) to finance major projects.

It represents the major source that governments and companies use to obtain **debt financing**. “The distinguishing characteristic of a bond is that the borrower (the issuer) enters into a legal agreement to compensate the lender (the lender, the bondholder)

through periodic interest payments in the form of coupons; and to repay the original sum (the principal) in full on a stipulated date, which is known as the bond's "maturity date."” (thau`bond`2011).

Important – Bond Market Price Variation

A bond can be traded freely in the marketplace. Its price varies continuously.

Table 4.4 summarizes the classification and key characteristics of different types of bonds.

Classification	Issued by	Characteristics	Examples
Treasury securities	Federal government	Backed by faith and credit of the federal government	Bills (≤ 1 year) Notes (2–10 years) Bonds (10–30 years)
Municipal	Local governments	Federal tax-exempt Issued against taxes received	General obligation Revenue Zero coupon Put
Mortgage ³	Corporation	Backed by specified assets or mortgage Low rate/low risk on first mortgage Foreclosure, if not repaid	First mortgage Second mortgage Equipment trust
Debenture	Corporation	Not backed by collateral, but by reputation of corporation Bond rate may “float” Higher interest rates and higher risks	Convertible Subordinated Junk or high yield

Table 4.4: Classification and Characteristics of Bonds

4.5.1 Characteristics of a Bond

- **Face value** (or **Par value**): the amount the borrower pays back on the stated maturity date.
- **Coupon amount**⁴ (or Bond interest): the amount paid periodically between the time money is borrowed and the time the face value is repaid.

⁴Before digital trading, bonds were physical paper certificates with small tear-off “coupons” attached. On the interest due date, the investor would detach a coupon and hand it to a bank to collect their cash payment.

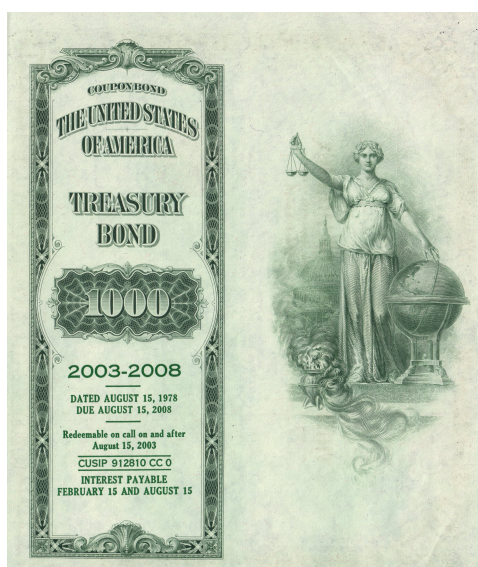


Figure 4.2: A United States treasury bond. Note that the bond clearly states the Face Value, issuance date, due date, and interest dates.

- **Bond coupon rate:** Annual interest rate.
- **Maturity Date:** The issuer must repay the Face Value on the maturity date.

Example 4.3. Consider the bond in Figure 4.2 has a coupon rate of 10%. The illustrated bond has a face value of \$1,000. The Bond interest or coupon amount is \$100 a year, paid in payments of \$50 on February 15 and August 15. On the maturity date (August 15, 2008), the bond issuer, that is the US government, **must** pay the bond holder (buyer) its face value of \$1,000.

Note – Callable Bond

In this particular example, the bond is a callable bond. This gives the issuer, (in this case, the U.S. government) the right, but not the obligation, to force an early end to the bond contract.

4.5.2 Cashflow

The cashflow for a bond investment is conventional, as shown in Figure 4.3. Because the cash flows of a bond are locked in by a legal contract, the only way for the open market to adjust to changing economic conditions is to change the upfront **Purchase Price** (P). To determine what a fair purchase price should be today, an investor must evaluate those fixed future cash flows against the interest rates currently available in the broader economy. This introduces a critical new characteristic: **the yield**.

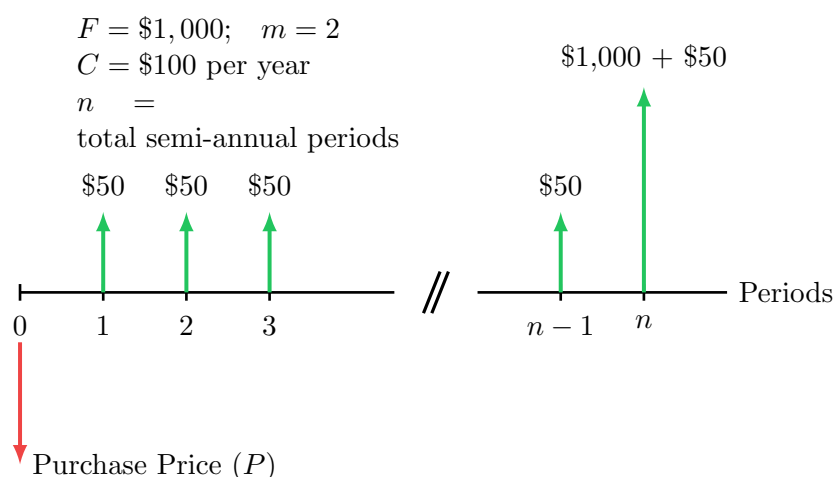


Figure 4.3: Cash flow diagram for the 10% U.S. Treasury bond from the **investor's perspective**. Notice that the final payment at period n includes both the face value and the final semi-annual coupon.

4.5.3 Yield to Maturity

Imagine buying a \$1,000 bond that pays \$50 a year (a 5% coupon). A year later, the economy shifts and new \$1,000 bonds are issued paying \$100 a year (a 10% return). If you try to sell your original bond for \$1,000, no one will buy it:

why buy yours for 50 a year when they can get 100 a year?

The present value of your bond is lower because the market has shifted. To sell your bond, you **must offer it at a “discount”** (e.g., selling it for 800). The buyer pays less upfront, which increases their **yield** (their actual rate of return) to match the new 10% market standard.

Definition – Yield to Maturity (YTM)

Yield to Maturity (YTM) is an estimate of the total rate of return anticipated to be earned by an investor who buys a bond at a given market price, holds it to maturity, and receives all interest payments and the capital redemption on schedule.

(thau'bond'2011)

Unlike coupon rate, the yield is a changing variable determined by the changes of the market:

- **Coupon Rate:** Fixed at issuance and determined by the borrower (issuer), it represents the **legal promise** of cash flow. It NEVER changes.
- **Yield to Maturity (YTM):** Variable, determined by the open market (investors), represents the current opportunity cost or required return. It changes every day based on the economy.

Consider a bond with a price of P and a face value F , making m coupon payments per year of C/m (with a total of n payments). See the respective cashflow in Figure 4.3.

The Yield to Maturity (YTM), denoted as λ , is the interest rate that equates the present worth of the bond's future cash flows to its current market price P . Using standard engineering economy equivalence factors, the price of the bond P is the sum of the present worth of the face value and the present worth of the uniform coupon series:

$$P = F(P/F, \lambda/m, n) + \frac{C}{m}(P/A, \lambda/m, n) \quad (4.2)$$

Where:

- $F(P/F, \lambda/m, n)$ brings the single Face Value payment back to year 0.
- $\frac{C}{m}(P/A, \lambda/m, n)$ brings the uniform series of coupon payments back to year 0.

By expanding these factors into their standard formulas (from Chapter 2), we get the full equation for the price of a bond:

$$P = F \left(1 + \frac{\lambda}{m} \right)^{-n} + \frac{C}{\lambda} \left(1 - \frac{1}{(1 + \lambda/m)^n} \right) \quad (4.3)$$

This formulation clearly shows that as the required yield λ increases, the denominator of the present worth factors increases. Therefore, the price of the bond P decreases as its yield increases. This explains why a high-yield bond commands a lower market price.

Note – Algebraic Derivation of the YTM Formula

For those interested in the mathematics, the uniform coupon series is a geometric sum:

$$\sum_{k=1}^n \frac{C/m}{(1 + \lambda/m)^k}$$

Using the mathematical identity for geometric series, $\sum_{k=1}^n \frac{1}{(1+x)^k} = \frac{1-(1+x)^{-n}}{x}$, and substituting $x = \lambda/m$, the summation algebraically reduces exactly to the (P/A) portion of Equation 4.3.

Important – Yield vs. Coupon Rate Relationship

Remember that the yield represents the broader market behaviour. When the yield is higher than the coupon rate, the present value of the bond becomes smaller than its face value because the market is providing higher rates of return.

This behaviour is illustrated in Figure 4.4 where the present value of the bond is exactly equal its par value when the yield and coupon rate are equal (10%).

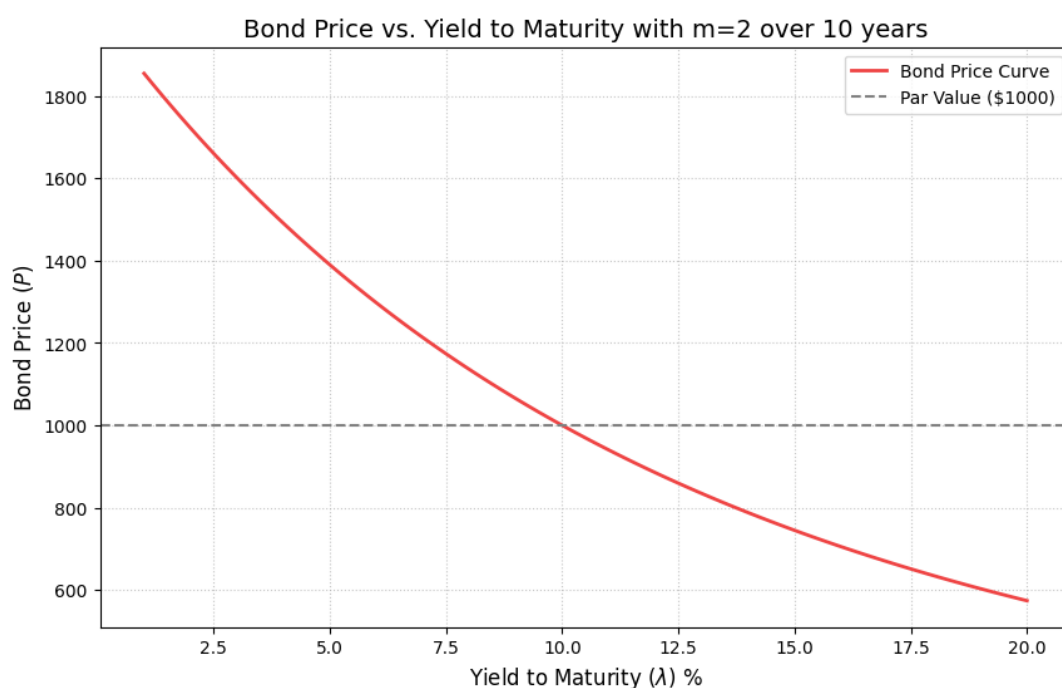


Figure 4.4: Figure highlighting how the price of a bond with face value of \$1,000 decreases as a function of the yield

Example 4.4. What is the price of a 10% (coupon, paid semi-annually), 30-year US Treasury bond with a yield of 4%? Assume a face value of 100. (Bond prices are typically quoted as percentage of face value.)

Solution In this example, $F = 100$, $C = 0.1 \times 100 = 10$, $\lambda = 4\%$, $m = 2$, and $n = 30 \times 2 = 60$.

$$P = F \left(1 + \frac{\lambda}{m} \right)^{-n} + \frac{C}{\lambda} \left(1 - \frac{1}{(1 + \lambda/m)^n} \right)$$

$$\begin{aligned}
&= 100 \left(1 + \frac{0.04}{2} \right)^{-60} + \frac{10}{0.04} \left(1 - \frac{1}{(1 + 0.04/2)^{60}} \right) \\
&= 100(0.3047) + \frac{10}{0.04}(0.69521) \\
&= 204.2725
\end{aligned}$$

Note – Solving via Excel

This could also be solved using Excel. It will be left to readers to try it out.

4.6 Time Varying Interest Rate

In practice, interest rates may vary from one period to the other. In particular, it is often **expected** that the interest rate will **increase** with time. This fact is reflected in the annual yield on long (maturity) bonds being higher than that on short bonds, as shown in Table 4.5.

Maturity Duration	Yield	Coupon Rate
3 Month	3.5%	-
6 Months	4%	-
12 Months	4.50%	-
36 Months	-	5.5%
60 Months	-	6%

Table 4.5: Sample yields rate for different maturity durations of Lebanese Treasury Bills (source:www.bdl.gov.lb). Note that Treasury Bills of duration less than or equal to a year do not pay coupons.

If the interest rates in periods, $1, \dots, n$ are $i_1, \dots, i_n$, then, the Future worth F of a Present amount P after n periods is:

$$F = P(1 + i_1)(1 + i_2) \dots (1 + i_n) = P \prod_{k=1}^n (1 + i_k)$$

The rates $i_1, \dots, i_n$ are known as **short rates**. The short rate i_n represents the expected 1-year rate after n years, and its value is estimated based on the yields of bonds (similar to the ones on Table 4.5).

Example 4.5. Justine leases a generator for private events. The net profit from the generator for each of the last 4 years has been decreasing, as shown below. Also shown are the annual rates of return on invested capital, which has been increasing.

Year	1	2	3	4
Net Profit	70,000	70,000	35,000	25,000
Annual Rate	7%	7%	9%	10%

Table 4.6: Net profit and annual rates of return for Justine's generator lease.

1-Draw the cashflow. 2-Determine the present worth P and 3-equivalent uniform series A of the net profit series. Take the annual variation of rates of return into account.

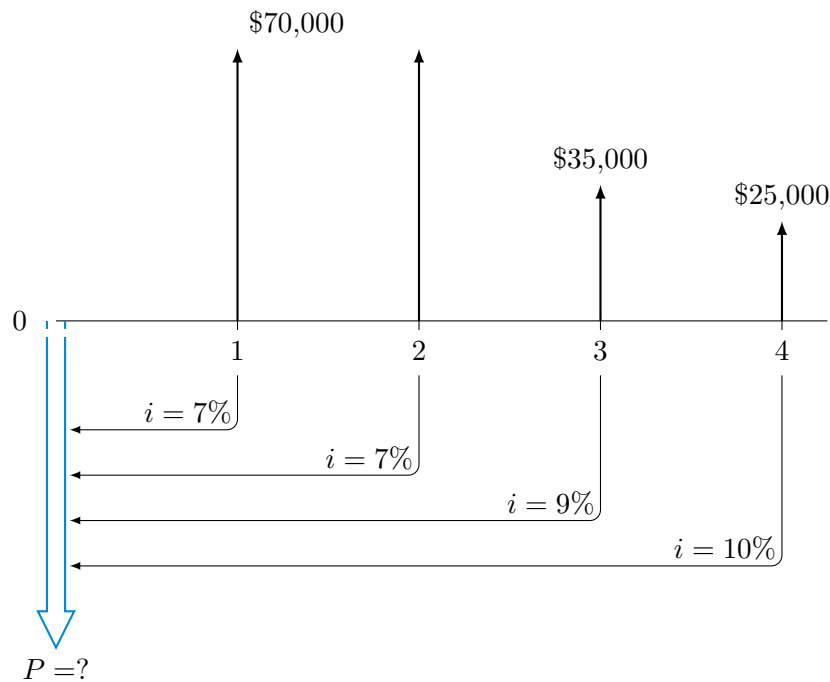


Figure 4.5: Cash flow diagram illustrating varying interest rates over time.

$$\begin{aligned}
 P &= \frac{70,000}{1 + 0.07} + \frac{70,000}{(1 + 0.07)^2} + \frac{35,000}{(1 + 0.07)^2(1 + 0.09)} + \frac{25,000}{(1 + 0.07)^2(1 + 0.09)(1 + 0.1)} \\
 &= \frac{70,000}{1.07} + \frac{70,000}{1.1449} + \frac{35,000}{1.247941} + \frac{25,000}{1.3727} \\
 &= 65,420.56 + 61,140.7 + 28,047.12 + 18,212.28 \\
 &= 172,820.6
 \end{aligned}$$

Alternatively, the value of P could be written in terms of the factors as:

$$\begin{aligned}
 P &= 70,000(P/A, 7\%, 2) + 35,000(P/F, 7\%, 2)(P/F, 9\%, 1) \\
 &\quad + 25,000(P/F, 7\%, 2)(P/F, 9\%, 1)(P/F, 10\%, 1)
 \end{aligned}$$

To find the equivalent **uniform** annual series, we rewrite the Present Value. An equivalent uniform series A is a constant annual cash flow that yields the exact same Present Value P as the original varying cash flows, when discounted using the **identical time-varying interest rates**:

$$P = \frac{A}{1 + 0.07} + \frac{A}{(1 + 0.07)^2} + \frac{A}{(1 + 0.07)^2(1 + 0.09)} + \frac{A}{(1 + 0.07)^2(1 + 0.09)(1 + 0.1)} \\ = A(0.9346 + 0.8734 + 0.80132 + 0.7285)$$

This will result in $A = \$51,777$ per year over the coming 4 years.

Important – Non-Uniform Interest Rates and A/P Factor

Applying the A/P factor formula we have seen in Chapter 2 requires a uniform interest rate across the periods. Hence, using it here is not possible.

For instance, if the average of the four annual rates, that is, 8.25% is used, the result is $A = 52,467$. This is a \$690 per year overestimate of the true equivalent annual net profit.

4.7 Summary

This chapter expanded the foundational equivalence concepts by introducing the critical distinction between nominal interest rates (the stated, uncompounded rate) and effective interest rates (the true, compounded rate). Furthermore, we explored how time-varying interest rates require a period-by-period analysis, highlighting the limitations of standard formulas when market conditions fluctuate.

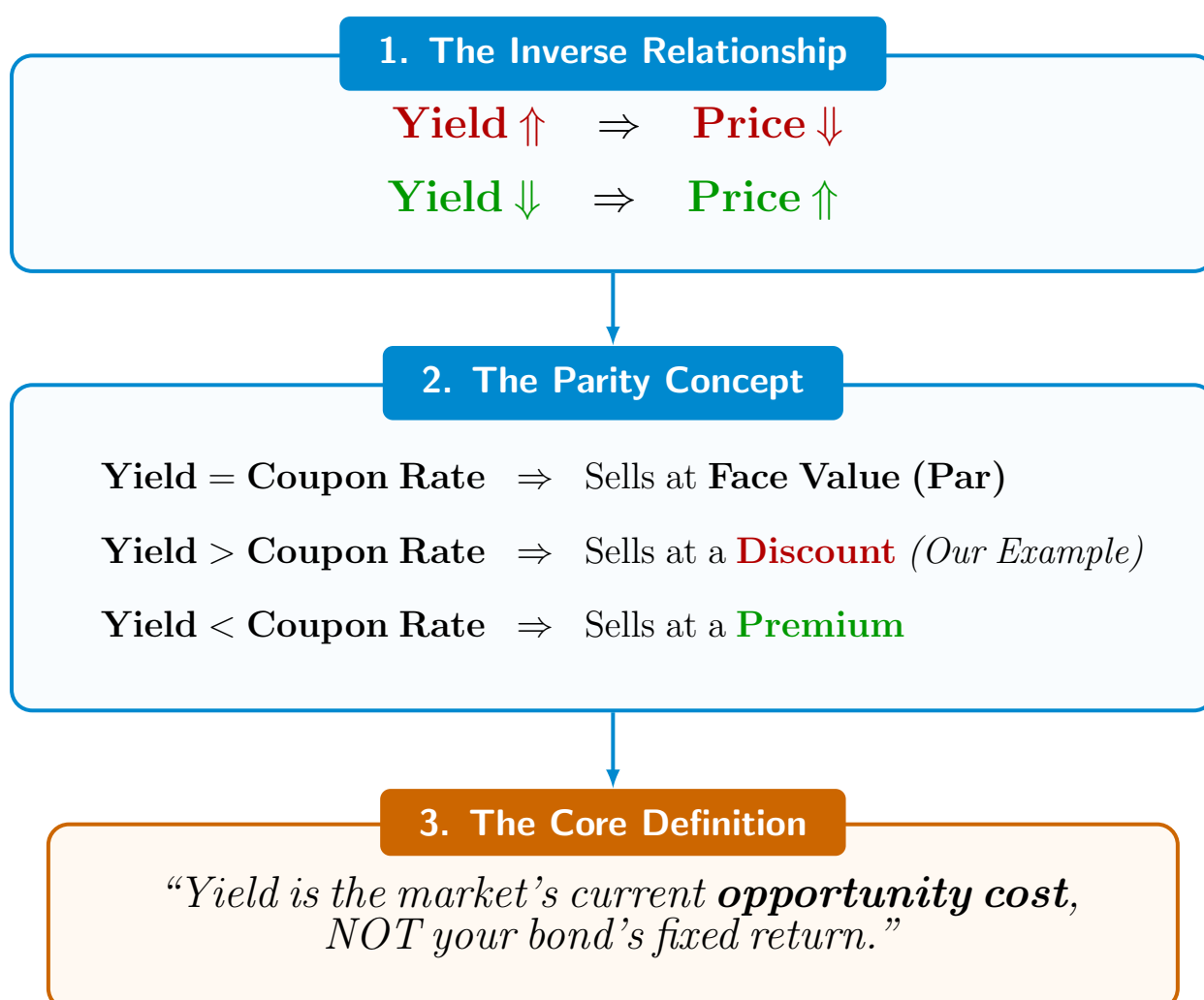


Figure 4.6: The three principles of bond pricing and yields.